

Comparing and Ordering Integers

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Integers include the **negative integers**, **zero**,
and **positive integers**.

A number line helps compare the values of integers.

An integer located farther to the **right** on the number line is **greater**.

An integer located farther to the **left** on the number line is **less**.

Use the symbols $>$, $<$, and $=$ to compare integers.

To **order integers**, arrange them from **least to greatest** or from **greatest to least**.

Negative numbers with larger absolute values are actually **smaller**. For example,

$$-9 < -4$$

Examples

Example 1

Compare: -7 _____ 3

Since -7 is to the left of 3 on the number line,

$$\boxed{-7 < 3}$$

Example 2

Compare: 5 _____ -2

Since 5 is to the right of -2 on the number line,

$$5 > -2$$

Example 3

Compare: -8 _____ -5

Since -8 is to the left of -5 on the number line,

$$\boxed{-8 < -5}$$

Example 4

Compare: 0 _____ -6

Since 0 is to the right of -6 on the number line,

$$0 > -6$$

Example 5

Compare: 4 _____ 4

The two integers have the same value.

$$4 = 4$$

Example 6

Arrange from least to greatest:

$$-2, 5, -7, 0$$

From left to right on the number line:

$$\boxed{-7, -2, 0, 5}$$

Example 7

Arrange from greatest to least:

$8, -1, 3, -6$

From greatest to least:

$8, 3, -1, -6$

Example 8

Arrange from least to greatest:

$-10, -3, 2, 7$

From least to greatest:

$-10, -3, 2, 7$

Example 9

Arrange from greatest to least:

$$0, -9, 6, -2$$

From greatest to least:

$$\boxed{6, 0, -2, -9}$$

Example 10

Arrange from least to greatest:

$9, -4, 1, -8, 0$

From least to greatest:

$-8, -4, 0, 1, 9$

Practice Exercises

Practice 1

Compare: -7 3

Practice 2

Compare: 5 -2

Practice 3

Compare: -8 -5

Practice 4

Compare: 0 -6

Practice 5

Compare: 4 4

Practice 6

Arrange from least to greatest:

$$-2, 5, -7, 0$$

Practice 7

Arrange from greatest to least:

$8, -1, 3, -6$

Practice 8

Arrange from least to greatest:

$-10, -3, 2, 7$

Practice 9

Arrange from greatest to least:

$0, -9, 6, -2$

Practice 10

Arrange from least to greatest:

$9, -4, 1, -8, 0$

Activities

Activity 1

Compare: -9 2

Activity 2

Compare: 6 - 5

Activity 3

Compare: -3 -11

Activity 4

Compare: 0 _____ 7

Activity 5

Compare: -1 -1

Activity 6

Arrange from least to greatest:

4, -6 , 1, -3

Activity 7

Arrange from greatest to least:

$-4, 9, 2, -8$

Activity 8

Arrange from least to greatest:

5, -9 , 0, -2

Activity 9

Arrange from greatest to least:

7, -5 , 3, 0

Activity 10

Arrange from least to greatest:

$-12, 8, -1, 4, 0$

Thank you!